

David Kreuter

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Senior Infrastructure & Network Automation Engineer

Infrastructure automation architect with 10+ years of experience building production-grade orchestration platforms, network provisioning systems, and API-driven lifecycle management at enterprise scale. Engineered a global provisioning framework managing 20,000+ virtual machines with automated IP allocation, firewall policy enforcement, and compute placement across geo-redundant data centers. Deep expertise in Python-based platform engineering, network security automation (firewall rule generation, DMZ provisioning, VLAN/subnet management), and Infrastructure as Code via Terraform and GitOps pipelines. Proven ability to design cloud-agnostic abstraction layers, high-availability architectures, and multi-tenant infrastructure in a regulated financial-services environment with 100% audit compliance.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Network Provisioning & Lifecycle Management:** Developed an intelligent IP-provisioning engine through InfoBlox API integrations with automated subnet allocation and network-resource lifecycle management, saving more than 1,000 engineering hours annually and resolving a multi-year infrastructure bottleneck.
- **Network Security Automation:** Automated Check Point firewall-rule generation and DMZ provisioning, integrating security-policy enforcement into the server build cycle to ensure day-zero network isolation, tenant separation, and a hardened security posture across all deployed infrastructure.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume, API-driven network and compute provisioning at scale.
- **Architectural Scale:** Engineered a global server-provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backup and compute placement, ensuring 100% RTO/RPO compliance and programmatic resource lifecycle management at build time.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate infrastructure deployment to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles, aligning with Infrastructure as Code best practices.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, designing an abstraction layer that enabled seamless multi-tenant deployment to Azure while maintaining a consistent provisioning interface transparent to end users.
- **Infrastructure Resilience:** Engineered and maintained a highly resilient infrastructure architecture spanning three identical, geo-redundant data centers with automated failover to ensure maximum uptime and disaster-recovery readiness.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow delivering fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and security-compliant deployments at scale across regulated financial infrastructure.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in "Read-Only" dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale-object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance across a distributed, multi-site environment.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid-cloud operations and cross-environment connectivity.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability.
- **Security Architecture:** Engineered a secure reverse-proxy solution to protect sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.

Network Engineer

- **Network Administration:** Managed complex firewall rules, switch ACLs, and routing policies for a large, distributed network spanning 12 airports and three satellite offices, ensuring tenant isolation and secure traffic segmentation.
- **Proactive Monitoring:** Developed service-aware alerting and telemetry frameworks for business-critical systems, reducing mean

time to resolution (MTTR) through proactive network observability and incident detection.

- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture and policy compliance.
- **Enterprise Mobility:** Onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geo-fencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics and network telemetry.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster-recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Languages: Python, Go (familiar), Bash, PowerShell, SQL, YAML, Javascript, PHP

Networking: BGP, DNS, VLAN/VXLAN, F5 Load Balancing, High Availability, VPN, VRRP, Linux Networking, Firewall Policy Management, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Automation/Orchestration: Terraform, REST API, GitOps, CI/CD Pipelines, Ansible Automation Platform, Middleware Development, Gitlab, Grafana, GraphQL, ServiceNow, Unit Testing

Infrastructure: Infrastructure as Code (IaC), Check Point Firewalls, VMware vSphere, Azure Cloud Services, OS Hardening (CIS/STIG), DBaaS, Nutanix AHV, Rubrik, Active Directory, NUMA Tuning

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

07/2025 - Present

Volunteer

Tift Nature Preserve

01/2025 - Present

Volunteer

716 Paws

01/2022 - Present

Thermal drone pilot for search and rescue missions.